# Hampus Serneke

Lausanne, Switzerland | +41 77 279 14 27 | hampus.serneke@epfl.ch | LinkedIn | GitHub | Personal

## EDUCATION

**École polytechnique fédérale de Lausanne, EPFL** **Lausanne, Switzerland**
*M.Sc. in Data Science*
- **Current coursework:** machine learning, measure theory, math of data, foundations of data science, law of AI
- **Upcoming coursework**: Foundation models and generative AI, modern NLP, visual intelligence

**Lund University** **Lund, Sweden**
*B.Sc. in Technology as part of the Engineering Mathematics program* Aug. 2022 – June 2026
- **Cumulative GPA:** 4.9/5.0
- Pursued a major in Computer Science and Engineering for two years before transferring to Engineering Mathematics
- **Thesis title:** Deep Learning for Rotation Averaging in Structure from Motion (GNNs in computer vision)
- **Coursework:** Mathematical statistics, algorithms and data structures, databases, matrix theory, stationary stochastic processes, applied mathematics (PDEs), numerical methods, functional programming, electromagnetic field theory

**Katedralskolan Lund** **Lund, Sweden**
*Natural Science High School program* Aug. 2018 – June 2021
- **GPA:** 22.5/22.5

## WORK & LEADERSHIP EXPERIENCE

**California Institute of Technology, Caltech** **Pasadena, USA**
*Summer Undergraduate Research Fellow* June 2026 – Aug. 2026
- Currently co-authoring a paper with our results. Pre-print will be on arXiv soon
- Conducted research in reinforcement learning and AI safety at the Advanced Mechanical Bipedal Experimental Robotics Lab under the supervision of Dr. Pol Mestres and Prof. Aaron Ames
- Created the first hardware implementation of an anytime safe reinforcement learning algorithm (RL-SGF), going from theorem to hardware
- Ran hardware experiments on G1 humanoids and Crazyflie quadrotors using policies trained in IsaacLab and mjlab
- Implemented a novel learning rate scheduler for fulfilling the safety requirements, increasing mean reward by 40%
- Obtained a sharper theoretical bound for the required learning rate to ensure safety
- Investigated the effects of relaxing the theory by introducing PPO-inspired estimates

**Lund University** **Lund, Sweden**
*Teaching Assistant in Mathematics, Faculty of Engineering* Aug. 2025 – Mar. 2026
- Led weekly exercise sessions where I taught calculus to a class of 30 engineering undergraduates

**Axis Communications** **Lund, Sweden**
*Machine Learning Intern, Core Technologies Analytics* June 2023 – May 2025
- Reduced bias in computer vision models using synthetic data
- Developed a pipeline for generating synthetic data in Unreal Engine
- Trained deep learning models for object detection and improved overall mAP, while reducing coverage bias by greatly improving performance on key object classes
- Implemented a novel noise reduction algorithm for pre-processing synthetic data in Python
- Streamlined the rendering pipeline in Unreal and refactored the codebase in C++

**Hemkodat** **Sweden**
*Coding Instructor* Sept. 2018 – Jan. 2023
- Travelled in Sweden during holidays and weekends to deliver courses in Python, Lua, and C# for adolescents

**The Computer Engineering Guild at LTH** **Lund, Sweden**
*Vice President* Jan. 2023 – Dec. 2023
- Led the board of a student guild with 1,000 members, chaired board meetings, and managed administration

## SKILLS, PROJECTS & OTHER

**Technical Skills:** Python (PyTorch), Java, C++, C#, Scala, Haskell, SQL, CI/CD, Git, ROS2, Isaac Lab

**Languages:** Swedish native, English C2 (IELTS Academic 8.5/9.0), Spanish A2, learning French

**Personal Projects:** Created a VR simulation of a student dorm in Unreal, developed an algorithm for solving Ricochet robots in Java, built an NLP model with Keras, developed regression models for biological experiments

**Open-Source:** Working on an implementation of a more stable geodesic distance loss function in the PyRoMA framework for geometric deep learning with PyTorch

**Awards:** Won the national qualifiers and received a bronze medal at the European Union Science Olympiad (2019)